

COL761 Assignment 2 Question 1

Determining Optimal k for K -Means

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1 Methodology and Selection of k

Given dataset $\{x_i\}_{i=1}^n \subset \mathbb{R}^d$ with $d > 3$, we minimise the k -means objective $J = \sum_{j=1}^k \sum_{x_i \in C_j} \|x_i - \mu_j\|^2$ for each $k \in \{1, \dots, 15\}$ using scikit-learn's `KMeans` with `n_init=20` restarts to mitigate sensitivity to initialisation. Dataset used: `dataset_num=2` for kerberos ID `me1221438`.

To select k we apply the **elbow method** with automatic detection: for each k , we compute the perpendicular distance of the point $(k, J(k))$ from the line joining $(1, J(1))$ to $(15, J(15))$. The k maximising this distance is the elbow. Figure 1 shows a sharp decrease in inertia up to $k = 6$, after which gains diminish. The automatic detector selects **$k = 6$** .

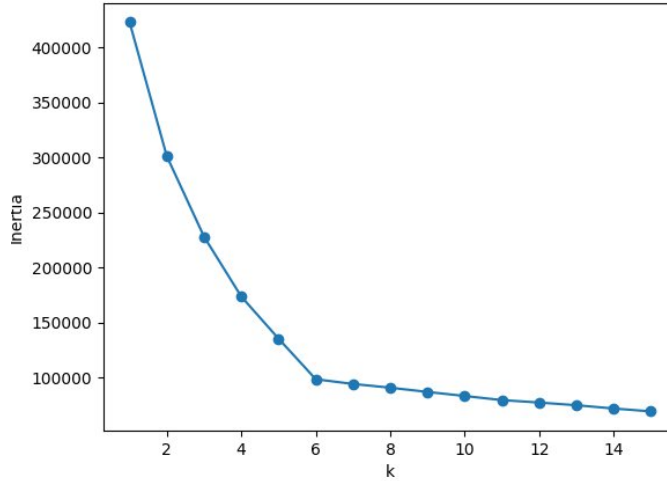


Figure 1: Elbow plot: inertia vs. k for `dataset_num=2`, kerberos ID `me1221438`.

2 Structural and Statistical Assumptions of k -Means

k -means implicitly assumes: (i) clusters are **convex and isotropic**—the Voronoi partition induced by centroids is optimal only when clusters are roughly spherical with similar variance; (ii) **equal cluster sizes** are preferred, since the squared-error objective tends to split large clusters and merge small ones; (iii) **Euclidean distance** is the appropriate dissimilarity measure, which presupposes that all features are on comparable scales; and (iv) the number of clusters k is **fixed a priori**, requiring an external model-selection criterion.

3 Limitations of the Elbow Method

The elbow method has several well-known weaknesses. **Subjectivity:** the “elbow” is often visually ambiguous, especially when inertia decreases smoothly without a clear kink. **Scale dependence:** the geometric distance heuristic is sensitive to the relative scaling of the k -axis and the inertia axis. **High-dimensional effects:** as d grows, pairwise distances concentrate [1], making inertia differences across k values smaller and the elbow harder to detect. **No statistical test:** unlike gap statistic or silhouette

analysis, the elbow method provides no confidence measure for the chosen k .

4 Methodological Improvements

Several alternatives offer more principled model selection:

Gap statistic [2]: compares $\log J(k)$ to its expected value under a reference null distribution (uniform), selecting k where the gap is maximised. This provides a formal hypothesis-test flavour.

Silhouette analysis: for each point, computes $(b - a) / \max(a, b)$ where a is mean intra-cluster distance and b is mean nearest-cluster distance. The average silhouette over all points peaks at the best k and additionally diagnoses cluster quality per point.

Information-theoretic criteria: BIC/AIC applied to a Gaussian mixture model (GMM) fit penalise model complexity explicitly, avoiding the need for visual inspection.

Stability-based selection: cluster the data on random subsamples for each k and measure consistency (e.g., via adjusted Rand index). The k yielding the most stable partition is selected.

5 When k -Means Fails and How to Detect It

Non-convex / elongated clusters. k -means splits crescent or ring-shaped clusters because Voronoi cells are convex. *Detection*: compute silhouette scores per cluster—low or negative values indicate points closer to a neighbouring centroid than their own. *Remedy*: spectral clustering or DBSCAN.

Clusters of varying density and size. The squared-error objective favours equal-mass partitions; a large diffuse cluster is often split while two small nearby clusters are merged. *Detection*: compare within-cluster variance across clusters; a high variance ratio signals this failure mode. *Remedy*: GMMs with full covariance, or density-based methods.

High-dimensional distance concentration. In high d , $\|x_i - x_j\| \approx \sqrt{d}\sigma$ for most pairs, so inter- and intra-cluster distances converge [1]. k -means may produce arbitrary partitions. *Detection*: compute the relative contrast $(\max\text{-dist} - \min\text{-dist}) / \min\text{-dist}$; values near zero indicate concentration. *Remedy*: apply PCA or random projection to reduce d before clustering, or use subspace clustering.

Sensitivity to initialisation. k -means converges to local optima. While `n_init=20` helps, pathological initialisations can persist. *Detection*: compare inertia across restarts; high variance signals instability. *Remedy*: k -means++ initialisation (default in scikit-learn) provably approximates $O(\log k)$ of optimal.

References

- [1] C. Aggarwal, A. Hinneburg, D. Keim, “On the surprising behavior of distance metrics in high dimensional space,” *ICDT*, 2001.
- [2] R. Tibshirani, G. Walther, T. Hastie, “Estimating the number of clusters in a data set via the gap statistic,” *JRSS-B*, 2001.